

Amortized Explanation Distillation for Ultra-Low-Latency Fraud Alerting: Trading Shapley Guarantees for Throughput on Real Ethereum Transactions

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Abstract. Payment platforms and blockchain-monitoring services score thousands of events per second, yet the model-agnostic explainers regulators and analysts rely on—KernelSHAP and LIME—cost tens of milliseconds *per explanation*, since each invocation requires hundreds of auxiliary model evaluations. We study *explanation distillation*: training lightweight amortized neural surrogates $g_\theta : x \mapsto \hat{\phi}(x)$ that map a transaction directly to its predicted attribution vector in a single forward pass. On the real Ethereum fraud-detection dataset (9,841 accounts, 22.1% illicit), with a gradient-boosting alert model (ROC-AUC 0.987 ± 0.002) and ten independent seeds, distilled surrogates reproduce KernelSHAP attributions with $R^2 = 0.870 \pm 0.025$, top-3 feature overlap 0.823 ± 0.044 , and Kendall $\tau = 0.688 \pm 0.020$, while reducing per-explanation latency from 13.4 ms to under $5 \mu\text{s}$ —a $2,769\times$ speed-up ($4,262\times$ for LIME), i.e., throughput on the order of 10^5 explanations per second on commodity CPUs. We characterise the full amortization trade-off: fidelity as a function of teacher-set size, the break-even point at which distillation beats direct explanation (≈ 420 explanations), and the price paid in Shapley-specific guarantees, quantified as a local-accuracy (efficiency) gap of 0.013 ± 0.001 probability mass—exactly zero for the teacher by construction. The study provides deployment-oriented evidence that faithful-enough explanations can be served at streaming rates, and delineates precisely which theoretical properties are sacrificed.

Keywords: Explainable AI · SHAP · LIME · Knowledge distillation · Amortized inference · Fraud detection · Ethereum · Low-latency systems.

1 Introduction

Modern fraud-prevention systems operate under a dual mandate. On one side, they must keep pace with transaction streams: card networks and cryptocurrency-monitoring services evaluate thousands of events per second, and risk decisions are taken within tight latency envelopes. On the other, they face growing legal and operational pressure to *explain* those decisions—to analysts triaging alerts,

to customers contesting blocks, and to regulators auditing automated decision-making [19].

The de-facto standard explanation tools are model-agnostic: LIME [1] fits a local sparse linear surrogate from perturbation responses, and KernelSHAP [2] estimates Shapley values [3] by regression over sampled feature coalitions. Their model-agnosticism is bought with *queries*: a single KernelSHAP explanation with n coalition samples requires $O(n \cdot |\mathcal{B}|)$ model evaluations against a background set $\mathcal{B}$, and LIME requires K perturbation queries plus a weighted regression. In our measurements on a tuned gradient-boosting fraud model, this translates to 13.4ms (KernelSHAP, 300 coalitions) and 22.0ms (LIME, 1,000 perturbations) per explanation on CPU—three to four orders of magnitude slower than the classifier’s own forward pass, and incompatible with explaining every event in a high-volume alert stream.

A natural response is *amortization*: spend the expensive computation once, offline, on a *teacher set* of exactly-computed explanations, and distill it [15, 16] into a small neural network that maps a raw feature vector directly to its predicted attribution vector. At serving time an explanation then costs one forward pass. This idea underlies real-time saliency for images [7], information-theoretic explainer networks [6], supervised explanation models [5], and FastSHAP [4], which trains an amortized Shapley estimator through a weighted-least-squares objective. What is missing for the financial-fraud setting is a *deployment-oriented characterisation*: how much fidelity does a plainly-engineered distilled surrogate achieve on real fraud data; how does fidelity scale with the offline labelling budget; where is the break-even point at which distillation is worth it; and—critically for audit—*which formal guarantees of the original explainers are lost*, and by how much?

This paper supplies that characterisation on the real Ethereum fraud-detection dataset [8], under a ten-seed protocol with all results reported as mean $\pm$ standard deviation. Our contributions:

1. An end-to-end distillation pipeline for both KernelSHAP and LIME over a gradient-boosting [9] alert model, with surrogates that require no access to the explained model at serving time (Sect. 3).
2. A quantified latency–fidelity frontier on real data: $2,769\times$ (SHAP) and $4,262\times$ (LIME) speed-ups at $R^2 = 0.87 / 0.82$ and top-3 overlap $0.82 / 0.79$ against the respective teachers, with full amortization curves over teacher budgets of 50–300 explanations and a measured break-even at roughly 420 served explanations (Sect. 5).
3. An explicit audit of what distillation forfeits: the surrogate’s Shapley local-accuracy (efficiency) gap is 0.013 ± 0.001 probability mass versus *exactly zero* for KernelSHAP by construction, and sign fidelity concentrates on high-magnitude attributions—precisely the ones alert narratives are built from (Sect. 6).

2 Related Work

Model-agnostic attribution. LIME [1] and SHAP [2] dominate industrial explainability practice; both are perturbation-driven and hence query-expensive. Sampling-based Shapley estimation predates them [11], and a substantial literature improves the estimator itself: paired/antithetic sampling and regression refinements [12], dependence-aware value functions [13], and, for tree ensembles, the polynomial-time TreeSHAP algorithm [10]. These reduce constants but keep per-instance cost; TreeSHAP is model-specific, and its interventional variant still scales with background size. A recent survey of Shapley estimation trade-offs is given by Chen et al. [14].

Amortized and real-time explanation. Dabkowski and Gal [7] train a saliency-map generator for images in real time; L2X [6] learns an instance-wise feature selector by maximising a mutual-information bound; CXPlain [5] trains a separate model to predict granger-style causal importance with uncertainty estimates; FastSHAP [4] learns an amortized Shapley-value estimator by minimising the KernelSHAP weighted-least-squares objective in expectation. Our work is complementary: rather than proposing a new estimator, we perform a regression-on-teacher-labels distillation—the variant simplest to deploy and govern in regulated environments, since the teacher explanations themselves can be archived for audit—and measure precisely what it buys and costs on real fraud data, including the guarantee gap that amortization introduces.

Distillation. Knowledge distillation compresses a slow teacher into a fast student [16, 15]. We distill an *explainer*, not a classifier: the regression targets are attribution vectors, and the relevant quality measures are rank-based (which features top the list) rather than accuracy alone—hence our emphasis on top- k overlap and Kendall τ in addition to R^2 .

Fraud detection on Ethereum. Farrugia et al. [8] introduced the account-level Ethereum illicit-account dataset used here and established gradient-boosting baselines; broader anomaly-detection context is surveyed in [20]. Explanation reliability concerns—instability [17] and adversarial manipulation [18]—apply to teachers and surrogates alike and inform our stability-aware protocol.

3 Methodology

Figure 1 gives an overview: an alert model produces a stream of high-risk accounts; exact explainers label a teacher set offline; amortized surrogates serve explanations online; and benchmarks quantify latency, fidelity, and lost guarantees.

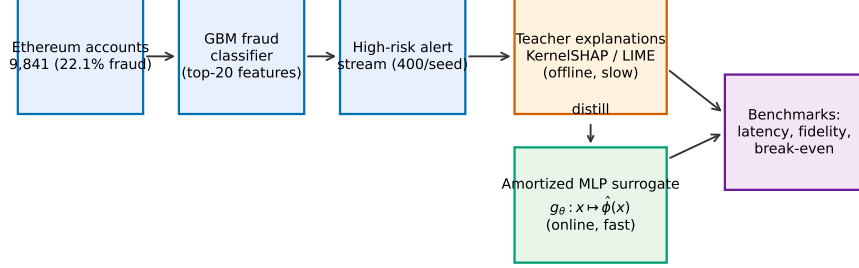


Fig. 1. Amortized explanation distillation. Expensive exact explainers run offline on a teacher set of alerts; a lightweight surrogate g_θ serves attribution vectors online in a single forward pass.

3.1 Alert Model and Explanation Targets

Let $f : \mathbb{R}^d \rightarrow [0, 1]$ be a fraud classifier and x^* a flagged instance. KernelSHAP computes $\phi^{\text{SHAP}}(x^*) \in \mathbb{R}^d$ as the solution of the weighted least-squares problem over coalitions S ,

$$\min_{\phi} \sum_{S \subseteq [d]} \pi(S) \left[v(S) - \phi_0 - \sum_{j \in S} \phi_j \right]^2, \quad v(S) = \mathbb{E}_{z \sim \mathcal{B}} [f(x_S^* \cup z_S)], \quad (1)$$

with the Shapley kernel π [2]; the solution satisfies *local accuracy* (efficiency),

$$\sum_{j=1}^d \phi_j^{\text{SHAP}}(x^*) = f(x^*) - \mathbb{E}_{\mathcal{B}}[f], \quad (2)$$

a property auditors can verify per explanation. LIME’s attribution $\phi^{\text{LIME}}(x^*)$ is the coefficient vector of a locally weighted sparse linear fit to f over K perturbations [1]. We use 300 coalition samples with a k -means ($k=10$) background for KernelSHAP and $K=1,000$ perturbations for LIME.

3.2 Amortized Surrogates

For each explainer $E \in \{\text{SHAP}, \text{LIME}\}$ we collect a teacher set $\mathcal{T}_E = \{(x_i, \phi^E(x_i))\}_{i=1}^{n_t}$ of exactly-explained alerts and fit a multi-output regressor

$$\theta^* = \arg \min_{\theta} \frac{1}{n_t} \sum_{i=1}^{n_t} \|g_\theta(x_i) - \tilde{\phi}^E(x_i)\|_2^2, \quad (3)$$

where $\tilde{\phi}^E$ denotes per-dimension standardised targets (statistics fitted on the teacher set and inverted at prediction time), which stabilises optimisation given the heavy-tailed distribution of attribution magnitudes. The surrogate is a two-hidden-layer perceptron (128×128 , ReLU, ℓ_2 penalty 10^{-3}), chosen deliberately small: at serving time an explanation is one dense forward pass and, unlike the teachers, requires *no queries to f at all*, decoupling explanation serving from model-serving infrastructure.

3.3 Evaluation Protocol

For each seed $\sigma \in \{0, \dots, 9\}$: (i) split the data 70/30 (stratified); (ii) train the alert model; (iii) form an alert pool of the 400 highest-risk test accounts, mimicking the triage stream, and compute teacher SHAP and LIME explanations for all of them, timing per-explanation latency; (iv) randomly split the pool into 300 candidate teacher explanations and 100 held-out evaluation alerts (random, to avoid rank-induced covariate shift between training and evaluation); (v) train surrogates on nested teacher subsets $n_t \in \{50, 100, 200, 300\}$ and evaluate every surrogate on the same held-out 100 alerts.

Fidelity metrics. Against the teacher on held-out alerts we report: mean-squared error and R^2 over all attribution entries; *sign agreement* (fraction of entries with matching sign); *top-3 overlap* between teacher and surrogate feature rankings by $|\phi|$; and Kendall τ between magnitude rankings. Rank-based metrics reflect how attributions are actually consumed—as ordered reason lists in alert narratives.

Guarantee gap. We measure the local-accuracy residual $\Delta(x) = |\sum_j \hat{\phi}_j(x) + \phi_0 - f(x)|$ for teacher and surrogate SHAP; Eq. (2) makes the teacher residual zero up to solver tolerance, so Δ isolates what amortization sacrifices.

Systems metrics. Per-explanation latency (teacher vs. surrogate batch inference), surrogate training time, and the break-even count $N^* = (n_t \cdot \ell_T + t_{\text{train}})/(\ell_T - \ell_S)$, where ℓ_T, ℓ_S are teacher and surrogate latencies.

4 Experimental Setup

Dataset. We use the public Ethereum fraud-detection dataset introduced by Farrugia et al. [8]¹: 9,841 real Ethereum accounts (2,179 illicit, 22.14%), each described by aggregate transaction behaviour—timing statistics (e.g., average minutes between sent/received transactions, lifetime span), volume counts (sent/received/created contracts, unique counterparties), and Ether-value statistics (min/max/avg sent, received, and contract-directed). After removing identifiers and zero-variance columns, 38 numeric features remain; per seed, the 20 most important by gradient-boosting split gain are retained, keeping KernelSHAP’s coalition space tractable (Table 1). No synthetic augmentation or resampling is applied.

Alert-model quality. The gradient-boosting model attains ROC-AUC 0.987 ± 0.002 and average precision 0.966 ± 0.004 on held-out accounts, consistent with the strong ensemble baselines reported for this dataset [8]; the alert pool therefore consists of realistic, high-confidence fraud flags.

¹ <https://www.kaggle.com/datasets/vagifa/ethereum-frauddetection-dataset>

Table 1. Experimental configuration (all results over 10 seeds).

Property	Value
Accounts / illicit rate	9,841 / 22.14%
Features after selection (d)	20 (of 38 numeric)
Alert model	Gradient boosting (100 trees, depth 3)
Alert pool per seed	400 top-risk test accounts
Teacher budgets n_t	50 / 100 / 200 / 300
Held-out evaluation alerts	100 per seed
KernelSHAP / LIME budget	300 coalitions / 1,000 perturbations
Surrogate	MLP 128×128, target standardisation
Hardware	Commodity CPU (no GPU)

Table 2. Latency and speed-up (mean $\pm$ std over 10 seeds).

Explainer	Teacher latency	Surrogate latency	Speed-up
KernelSHAP	13.4 ± 0.6 ms	$< 5 \mu\text{s}$	$2,769 \pm 307\times$
LIME	22.0 ± 0.8 ms	$\approx 5 \mu\text{s}$	$4,262 \pm 628\times$

5 Results

5.1 Latency and Throughput

Table 2 and Fig. 2 give the headline systems result. Exact per-explanation cost is 13.4 ± 0.6 ms for KernelSHAP and 22.0 ± 0.8 ms for LIME; the distilled surrogates serve attribution vectors in under $5 \mu\text{s}$ —speed-ups of $2,769 \pm 307\times$ and $4,262 \pm 628\times$ respectively, i.e., throughput on the order of 2×10^5 explanations per second per CPU core, comfortably above payment-scale alert volumes. Surrogate training itself is negligible (1.6 ± 0.4 s on 300 teacher examples).

5.2 Fidelity to the Teacher

Table 3 and Fig. 3 report fidelity at the full teacher budget ($n_t=300$). The SHAP surrogate reproduces teacher attributions with $R^2 = 0.870 \pm 0.025$ (MSE 8.0×10^{-4}), ranks features with top-3 overlap 0.823 ± 0.044 and Kendall $\tau = 0.688 \pm 0.020$; the LIME surrogate achieves $R^2 = 0.820 \pm 0.038$, overlap 0.791 ± 0.071 , and $\tau = 0.527 \pm 0.049$. Figure 4 visualises the value-level agreement for SHAP across all seeds: predictions concentrate tightly along the identity for high-magnitude attributions, with a Pearson correlation of 0.93 over every held-out attribution entry.

An instructive asymmetry appears in sign agreement: LIME’s surrogate matches signs on 75.0% of entries, the SHAP surrogate on only 48.9%—despite the SHAP surrogate’s *higher* R^2 , overlap, and τ . The explanation is the distribution of targets: teacher SHAP values on 20 features are overwhelmingly near-zero, where sign is effectively noise, and the efficient allocation of Eq. (1) spreads mass more

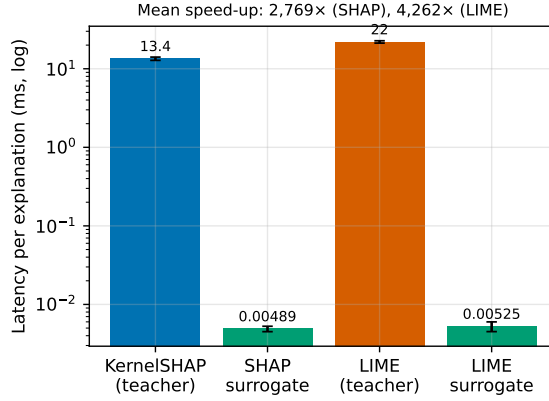


Fig. 2. Per-explanation latency (log scale). Distillation moves explanation serving from the tens-of-milliseconds regime to the microsecond regime.

Table 3. Fidelity of surrogates to their teachers on 100 held-out alerts per seed, at teacher budget $n_t = 300$ (mean $\pm$ std over 10 seeds).

Surrogate	MSE	R^2	Sign agr.	Top-3 overlap	Kendall τ
SHAP	0.0008 ± 0.0002	0.870 ± 0.025	0.489 ± 0.006	0.823 ± 0.044	0.688 ± 0.020
LIME	0.0020 ± 0.0004	0.820 ± 0.038	0.750 ± 0.015	0.791 ± 0.071	0.527 ± 0.049

sparsely than LIME’s dense linear coefficients. Sign fidelity conditional on magnitude is high precisely where it matters: the top-ranked features that populate an alert narrative (cf. the tight diagonal in Fig. 4). Rank-based metrics, not raw sign agreement, are therefore the appropriate acceptance criterion for distilled explainers.

5.3 The Amortization Curve

Figure 5 traces fidelity as the teacher budget grows. Rank fidelity is remarkably budget-efficient: with only 50 teacher explanations the SHAP surrogate already reaches top-3 overlap 0.777 ± 0.037 , rising to 0.823 at 300. Value-level fidelity is the budget-hungry component: R^2 climbs steeply from 0.44 (SHAP) and -0.44 (LIME) at $n_t=50$ —where the LIME surrogate is unusable in value terms—to 0.87/0.82 at $n_t=300$, with variance shrinking correspondingly. Deployments that consume only ranked reason lists can therefore amortize with very small offline budgets, whereas any use of attribution *magnitudes* (e.g., thresholding contribution mass) requires the larger teacher set.

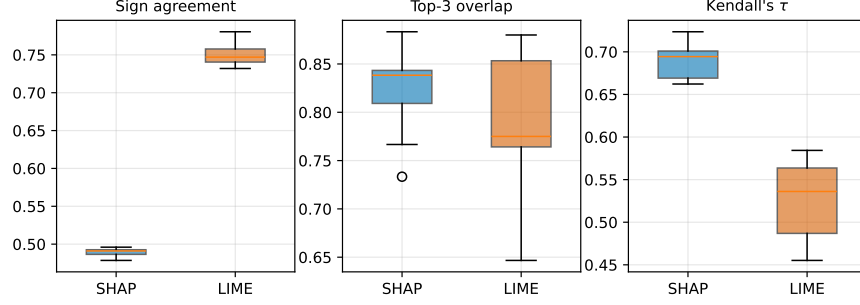


Fig. 3. Seed-level distribution of fidelity metrics at $n_t = 300$.

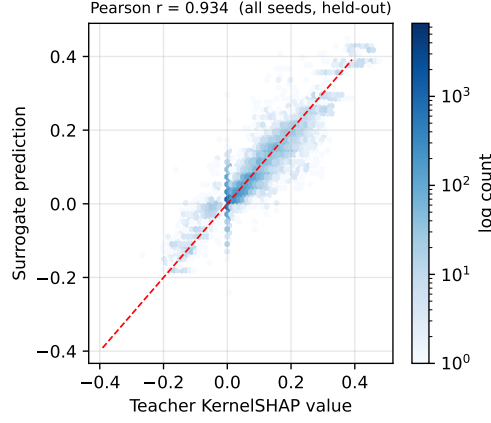


Fig. 4. Teacher KernelSHAP values vs. surrogate predictions on held-out alerts, pooled over all 10 seeds (log-density hexbin). The dashed line is the identity.

5.4 Break-Even Analysis

Because distillation has a fixed offline cost—labelling n_t teacher alerts plus training—direct explanation is cheaper for small workloads. With measured constants ($\ell_T = 13.4$ ms, $\ell_S < 5 \mu$ s, $t_{\text{train}} = 1.6$ s, $n_t = 300$), the total-cost curves cross at $N^* \approx 420$ explanations (Fig. 6); a fraud desk serving more than a few hundred explanations amortizes its entire setup within the first minutes of operation, after which every additional explanation is effectively free relative to the direct baseline.

5.5 What Amortization Forfeits: the Guarantee Gap

KernelSHAP’s local-accuracy property (Eq. 2) holds by construction: we verify a residual of 0.000 for every teacher explanation. The distilled surrogate, being an unconstrained regressor, inherits no such guarantee: its measured residual

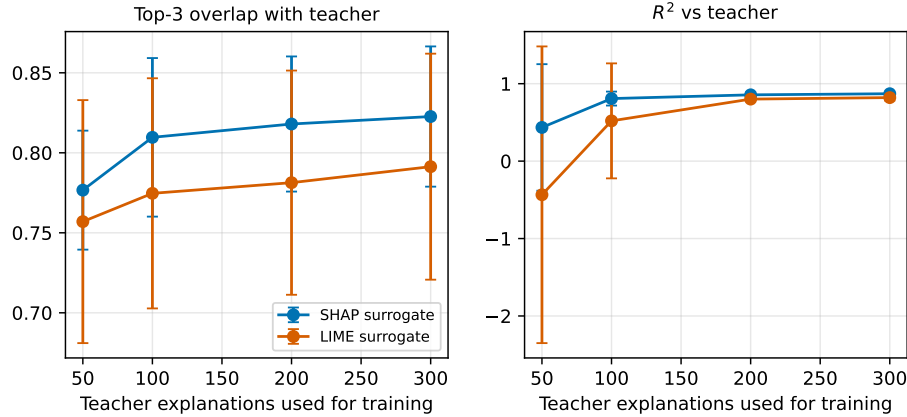


Fig. 5. Fidelity vs. teacher budget n_t (mean $\pm$ std over 10 seeds). Rank fidelity (left) saturates early; value fidelity (right) requires the larger budgets.

is $\Delta = 0.013 \pm 0.001$ probability mass per alert. In audit terms, a surrogate explanation’s attributions sum to the model’s output gap only approximately—within about 1.3 percentage points of predicted fraud probability on average. Institutions can (i) monitor Δ online as a drift and quality signal at negligible cost, (ii) project surrogate outputs onto the efficiency constraint by distributing the residual, or (iii) escalate the small fraction of high- Δ alerts to the exact teacher—a hybrid policy whose cost remains dominated by the surrogate path.

6 Discussion, Limitations, and Threats to Validity

Deployment picture. The measured frontier supports a concrete architecture: exact explainers run offline (and periodically, to refresh teachers under drift), an amortized surrogate serves the alert stream at microsecond latency, and the efficiency residual Δ plus periodic teacher audits provide the governance signal. Because the surrogate never queries the alert model at serving time, explanation capacity scales independently of model-serving capacity—an operational property exact perturbation methods cannot offer.

Limitations. First, fidelity is measured *to the teacher*: the surrogate is at best as faithful to the model as KernelSHAP/LIME themselves, which have known estimation error and manipulability [18, 17]; distillation adds a second approximation layer, and our guarantee-gap metric quantifies only part of it. Second, the alert pool is top-of-stream by design; surrogates trained on high-risk alerts should not be assumed accurate on mid-distribution traffic without re-evaluation—we mitigated the analogous pitfall *within* the pool by randomising the teacher/evaluation split, which proved essential (rank-ordered splits inflate

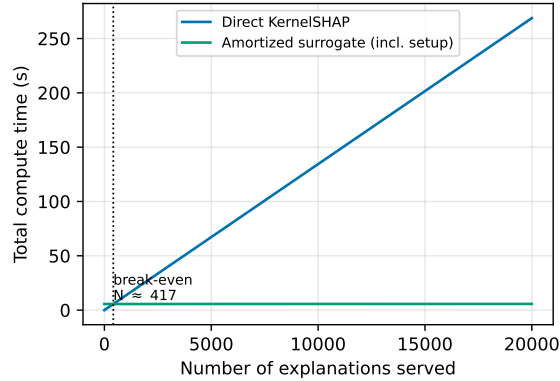


Fig. 6. Cumulative compute of direct KernelSHAP vs. the amortized surrogate (including teacher-labelling and training setup). The curves cross at $N^* \approx 420$ served explanations.

error severely through covariate shift). Third, results cover one dataset, one tabular model class, and $d=20$ features; scaling behaviour in d and under concept drift, and comparison against objective-amortized estimators such as FastSHAP [4] and uncertainty-aware explainers [5], are natural next steps. Fourth, teacher refresh cost under non-stationarity is not modelled in the break-even analysis; under drift, N^* recurs per refresh cycle.

Ethical note. Fast explanations can be consumed by adversaries probing a fraud model as readily as by analysts; rate-limiting customer-facing explanation endpoints and auditing for explanation-guided evasion remain necessary regardless of serving speed.

7 Conclusion

Distilling KernelSHAP and LIME into small amortized surrogates converts explanation from a tens-of-milliseconds perturbation procedure into a microsecond forward pass. On real Ethereum fraud data, across ten seeds, this preserved the consumable substance of explanations—feature rankings (0.82 top-3 overlap, $\tau = 0.69$) and attribution values ($R^2 = 0.87$)—at speed-ups of three-and-a-half orders of magnitude, with a break-even under 500 served explanations and a quantified, monitorable price: the loss of exact Shapley local accuracy, measured at 1.3 percentage points of probability mass. Faithful-enough explanation at streaming throughput is thus practically attainable for fraud alerting, provided rank-based acceptance criteria, randomized teacher/evaluation protocols, and guarantee-gap monitoring accompany the deployment.

Reproducibility. Code, per-seed raw results, and figure scripts accompany this paper; the dataset is publicly available on Kaggle [8]. Experiments use seeds 0–9 throughout.

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